

UTSAVKUMAR VAGHANI

Senior Backend Engineer | Distributed Systems | Payments and Production GenAI

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Summary

Backend engineer with 5+ years in fintech (Go, Node.js, Python). Built payments infrastructure moving **USD 1.5M+/yr** across four gateways, investment alerting for **5M users**, and the order-management core of a multi-market wealth platform. At Razorpay, a production GenAI system I led **removed INR 50L/month** of support cost while **auto-resolving 300–400 tickets/day**.

Experience

Keenai Global

Aug 2025 – Present

Senior Associate, Product Engineering (Backend Engineer)

Bangalore, India

- Built the **order-management service** for all instrument types (equities, ETFs, MFs, options, FCNs, hedge funds) across **US, Singapore and more** with **Temporal-orchestrated** workflows; **cut list-orders latency by ~10s**.
- Built the custodian **execution-report pipeline** on SQS pub/sub (**per-order FIFO**, **dedupe** on execution IDs, batching, **Redlock** locks) and the **org-wide domain-events service** with **Kafka (MSK, IAM)**, FIFO partition keys and DLQs.
- Enabled **real-time portfolio data**: OMS → portfolio-service sync on order fulfilment, **chained jobs with Postgres debounce** and **cross-account EventBridge**; reconciled against bank holdings.
- Led the **portfolio revamp** replacing a **third-party analytics vendor with in-house services** — overview, holdings, allocation, performance (TWRR, XIRR, AUM) — from spec to go-live; **own the portfolio module** for the wealth platform.
- Built a **price-hierarchy engine** picking each instrument's valuation price across market-data sources; **cut recon breaks 20%**.
- Built **portfolio versioning (business-day + intraday)** over **100K+ transactions per family office**: SFO/entity-level **Ops dashboard** (versioned-vs-live views, **audit logging**), **auto-versioning on successful reconciliation**.
- Built the **reporting module end-to-end**: async **Handlebars-templated** generation of transaction and holdings reports, plus a **drag-and-drop template builder** with **per-tenant branding** for RM, IC and client reports.
- Built **Skipper**, a pluggable **AI incident-triage service**: polls **CloudWatch** error logs, applies **per-service alert rules** (spike thresholds, cooldowns, known-signature dedupe), runs an **agentic Claude RCA** over logs, git history and read-only prod data, and posts **root cause, temp fix and permanent fix to Slack**; a **critic pass and confidence gate** suppress low-evidence RCAs.
- Built the reporting service's **code-quality system**: a **37-assertion conventions rubric**, **blocking CI gates** (Prettier, type-aware ESLint with import-boundary rules, tsc) and an **advisory Claude PR reviewer** that checks **architecture and correctness** against the rubric, cites the violated assertion ID and posts inline comments before human review.
- Stack: Node.js (TypeScript), Java, Python, PostgreSQL, DynamoDB, Redis, Kafka (MSK), SQS, EventBridge, Temporal, Lambda

Razorpay

Sep 2024 – Aug 2025

Senior Software Engineer

Bangalore, India

- Built **Yogi**, the GenAI ticket-resolution platform — intent classification, RAG over SOPs, tool-calls into internal systems, confidence-gated auto-resolution — **auto-resolving 300–400 tickets/day** and cutting support cost by **INR 50L/month**.
- Built a **RAG + LLM merchant-onboarding assistant** with SOP-driven flows reaching **70% containment**; instrumented containment, accuracy, latency and cost-per-ticket metrics for safe rollout.
- Led and mentored 5 engineers**; owned availability, on-call and runbooks; took the legacy central service **from weekly to on-demand deploys**.

Smallcase

Mar 2022 – Sep 2024

SDE I → SDE II

Bangalore, India

- Built a **payments PaaS from scratch** across **Razorpay, Juspay, Apple IAP and Zerodha (USD 1.5M+/yr)**: one gateway interface, payment state machine, idempotent writes, webhook dedupe, settlement reconciliation (Go, NestJS, MongoDB).
- Built **investment alerting from scratch** for **5M users (1M active)**: pre-market corporate-action, intraday price and end-of-day alerts.
- Made legacy **Zerodha (Kite) transaction flows fault-tolerant** (idempotent retries, recon); **revamped the offer engine**.

Zopsmart

Jan 2021 – Feb 2022

SDE Intern → SDE I

Bangalore, India

- Built a **Go app framework** (web and CLI) with generic pub/sub, metrics and AWS interfaces; **adopted org-wide**.
- Built **CSP** (credential service provider) for authenticated inter-service communication; end-to-end APIs for a marketplace app.

Projects

- docdeck** (TypeScript, MIT): markdown → step-through decks; mermaid, speaker view, PDF export.
- Brahmos** (Python, in progress): Indian-markets algo backtester; YAML strategy IR, honest cost/slippage/margin model.

Skills

- Languages**: Go (Golang), JavaScript/TypeScript (Node.js), Python
- Backend**: gRPC, REST, NestJS, Gofiber, FastAPI, Celery, Kafka, SNS/SQS, Temporal, event-driven
- GenAI**: Claude and OpenAI APIs, agentic tool-use, RAG, MCP, n8n, AI code review in CI, LLM evals and observability
- Data and infra**: PostgreSQL, MongoDB, DynamoDB, MySQL, Redis, AWS (Lambda, SNS, SQS, S3), Docker, Prometheus, CI/CD

Education and Certifications

- BE, Computer Engineering**, G H Patel College of Engg. & Tech., 2017–2021, CGPA 8.22/10 · **AWS Certified Cloud Practitioner** · AWS Partner Accreditation